

# Dr. Víctor Manuel Chávez Pérez

## Professor–Researcher · Applied Physics, Data Science & Science Education

Centro Universitario de Tlaquepaque (Tlaquepaque University Center), Universidad de Guadalajara · Tecnológico de Monterrey, Guadalajara Campus · Zapopan, Jalisco · Mexico · [victor.chavez@academicos.udg.mx](mailto:victor.chavez@academicos.udg.mx) · ORCID: 0000-0002-6761-0730 · [victorchavezudg.github.io/personalweb](https://victorchavezudg.github.io/personalweb)

[Google Scholar](#) · [ResearchGate](#) · [LinkedIn](#)

SNII · Candidate (SECIHTI) 2025–2029



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## PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), heat waves and their response under climate-change scenarios using models such as CMIP5 and WACCM.

Beyond basic research, I bring physics to real-world problems: I am developing a flood early-warning system for the Guadalajara Metropolitan Area that integrates weather radar, hydrological runoff modeling over high-resolution LiDAR terrain and remote sensing. My work also spans physical oceanography, remote sensing and applied optics.

I am also interested in how generative AI can transform science education: I design learning assistants and teaching materials that I evaluate with students themselves. Alongside this, I co-founded the Tlakakini Science & Technology Club, devoted to science outreach for children. I believe the best science is the kind that gets shared.

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## RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · Data science · High-performance computing · Science education & outreach · Generative AI in science education · Machine Learning applied to climate & atmospheric data · Student perception of AI-based learning tools

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## EDUCATION

- 2017 – 2024**    **PhD in Applied Physics**  
Universidade de Vigo · Spain  
Thesis: Sensitivity of sudden stratospheric warmings to simulation and analysis conditions in models.
- 2011 – 2012**    **MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change**  
Universidade de Vigo · Spain  
Temporal and spatial variability of the Minho river plume (MODIS 250 m). Top honors.
- 2009 – 2011**    **MSc in Hydrometeorology — Meteorology & Physical Oceanography**  
Universidad de Guadalajara · Mexico  
Seasonality and anomalies of oceanographic variables in the Gulf of Mexico via remote sensing.
- 2000 – 2008**    **BSc in Physics**  
Universidad de Guadalajara · Mexico  
Thesis: Ocean and coastal meteorology of Nayarit (San Blas, 1999–2002). Grade 95/100.

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## EXPERIENCE

- 2026 – present**    **Adjunct Professor ("Profesor de Cátedra")**  
Tecnológico de Monterrey · Campus Guadalajara · Zapopan, Jalisco · Mexico  
Course unit F1025B, The Laws of Motion and Conservation, team-taught by two instructors — one for theory and one for the challenge. Two theory groups and one challenge group.  
**Courses:** The Laws of Motion and Conservation (Theory) ×2 · The Laws of Motion and Conservation (Challenge)
- 2025 – 2026**    **Lecturer**  
Centro Universitario Bonpland y Humboldt · Online  
PhD in Aerospace Engineering: Research Methodology and Fluid Mechanics.  
**Courses:** Research Methodology ×2 · Fluid Mechanics
- 2024 – present**    **Lecturer "B"**  
Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico  
Department of Basic Sciences. Chair of the Basic Sciences Academic Board.  
**Courses:** Calculus ×2 · Advanced Calculus ×2 · Numerical Methods ×2 · Probability · Electromagnetism Lab · Discrete Mathematics

## 2024 – present External Collaborating Researcher

Universidade de Vigo · EPhysLab · Ourense, Spain

Doctoral research on sudden stratospheric warmings in CMIP5 and WACCM models; statistical analysis of large climate datasets.

## 2019 – 2026 Adjunct Professor ("Profesor de Asignatura B")

Universidad Tecnológica de Jalisco · Guadalajara, Mexico

Mechatronics Division — Basic Sciences.

**Courses:** Engineering Physics ×10 · Probability & Statistics ×8 · Engineering Mathematics II ×6 · Differential Calculus ×5 · Integral Calculus ×3 · Engineering Mathematics I ×2 · Multivariable Calculus · Applied Engineering Statistics · Thermodynamics

## 2019 Faculty Lecturer

Tecmilenio · Campus Guadalajara · Guadalajara, Mexico

High school: Matter & Energy, Integral Calculus, Science & Technology.

**Courses:** Integral Calculus ×2 · Differential Calculus · Mathematics (Language of Science) · Matter & Energy · Science & Technology

## 2019 Physics Instructor

UNITEC · Campus Guadalajara · Guadalajara, Mexico

Engineering programs: Statics.

**Courses:** Statics

## 2015 Predoctoral mobility stay

Universidad Pablo de Olavide · Seville, Spain

Short research stay at R&D centers; results presented at international conferences.

## 2013 – 2017 Predoctoral Researcher (FPI Fellowship)

Universidade de Vigo · EPhysLab · Ourense, Spain

Statistical analysis of large-volume data for climate-change research with general circulation models.

## 2013 – 2014 Predoctoral research stay

Universidad de Guadalajara · Guadalajara, Mexico

Academic research stay on climate-change analysis.

## 2010 – 2012 Physics & Math Instructor

Centro de Enseñanza Técnica Industrial (CETI Colomos) · Guadalajara, Mexico

Engineering programs.

**Courses:** Differential & Integral Calculus ×2 · Statics ×2 · Dynamics ×2 · Mathematics ×2

## 2010 – 2011 Physics & Math Instructor

Universidad de Especialidades · Guadalajara, Mexico

Mechanics, Introduction to Physics and Electromagnetism.

**Courses:** Mechanics · Intro to Physics · Electromagnetism

## 2008 – 2011 Professor–Researcher

Instituto Tecnológico Superior de Zapopan · Zapopan, Mexico

Mathematics for engineering; parallel computing applied to oceanography.

**Courses:** Physics (Mechanics, E&M, Thermodynamics) ×11 · Mathematics & Calculus (Differential, Integral, Vector, ODEs) ×17 · Dynamics

## 2006 – 2007 Research Assistant

Universidad de Guadalajara · CUCEI, Departamento de Física · Guadalajara, Jalisco · Mexico

"Oceanografía de la Costa de Nayarit" project (CONACYT-SEP-2003-C02-44870F). PAY and PITAI programmes.

## 2005 – 2006 Physics, Chemistry and Mathematics Teacher

Centro Universitario UTEG · Guadalajara, Jalisco · Mexico

Intensive semi-schooled high school programme (BIS).

## PUBLICATIONS

### 2025 Observación Astronómica en México: Pasando de lo Visible a lo Invisible

Almaguer-Gómez C., Medina J. F., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada* 58(1):15 · DOI: [10.7149/OPA.58.1.51200](https://doi.org/10.7149/OPA.58.1.51200)

### 2025 Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations

Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 16(5) · DOI: [10.3390/atmos16050628](https://doi.org/10.3390/atmos16050628)

### 2025 Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios

Chávez-Pérez V. M., et al. · *Climate* 13(10) · DOI: [10.3390/cli13100207](https://doi.org/10.3390/cli13100207)

- 2023**      **Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría**  
Palacios Hernández E., Chávez Pérez V. M., Sierra Romero L. L. · *Generis Publishing* · ISBN: 979-8-88676-642-4
- 2022**      **Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings**  
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 13(4) · DOI: [10.3390/atmos13040546](https://doi.org/10.3390/atmos13040546)
- 2018**      **The climatology of dust events over the European continent using data of the BSC-DREAM8b model**  
Mandija F., et al. · *Atmospheric Research* 209:144–162 · DOI: [10.1016/j.atmosres.2018.03.006](https://doi.org/10.1016/j.atmosres.2018.03.006)
- In preparation**
- 2026**      Science Beyond the Classroom: Impact Assessment of a Monthly Science Club During National Non-School Days in Mexico  
Almaguer-Gómez C., Chávez-Pérez V. M., et al.
- 2026**      Student Perception of a Generative AI-Based Calculus Learning Assistant: A Mixed-Methods Study  
Chávez-Pérez V. M., et al.
- 2026**      Assessment of the lightning activity response to aerosol optical depth  
Mandija F., Chávez-Pérez V. M.

## CONFERENCES & TALKS

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- 2022**      **65th Mexican National Physics Congress**  
2–7 oct 2022 · Zacatecas, Mexico  
• Analysis of the occurrence frequency of Southern Hemisphere stratospheric sudden warmings in high-top CMIP5 models  
• Optical characterisation of transparent 3D-printing resin
- 2020**      **63rd Mexican National Physics Congress**  
4–9 oct 2020 · Online  
• Significance of basic SSW characteristics when comparing CMIP5 models with reanalysis data over pre-satellite and post-satellite periods  
• Impact on the occurrence frequency of major stratospheric sudden warmings under different climate change scenarios using CMIP5
- 2018**      **American Geophysical Union (AGU) Fall Meeting 2018**  
10–14 dic 2018 · Washington, USA  
• Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Stratospheric Sudden Warmings
- 2018**      **Mexican Geophysical Union Annual Meeting**  
28 oct–2 nov 2018 · Puerto Vallarta, Mexico  
• Changes in the climatology of major stratospheric sudden warmings using WACCM with high vertical resolution  
• Assessment of the ability to reproduce major stratospheric sudden warmings in the CMIP5 historical dataset  
• Major stratospheric sudden warmings under climate change scenarios with CMIP5
- 2017**      **European Geosciences Union (EGU) General Assembly 2017**  
23–28 abr 2017 · Vienna, Austria  
• Estimation of the variability of dust aerosol optical depth over several European regions based on forecast model data
- 2016**      **59th Mexican National Physics Congress (SMF)**  
2–7 oct 2016 · León, Mexico  
• Stratospheric teleconnection and stratospheric sudden warmings  
• Characteristic patterns in the stratospheric variables of the Quasi-Biennial Oscillation  
• Stratospheric anomalies before and after stratospheric sudden warmings for the different phases of the Quasi-Biennial Oscillation
- 2016**      **SPARC Workshop SHARP 2016 — Stratospheric Change and its Role for Climate Prediction**  
11–15 jul 2016 · Berlin, Germany  
• Changes in the stratosphere before and after SSW events in the different phases of the Equatorial QBO  
• Differences in the detection and classification of the Stratospheric Sudden Warming over the three reanalyses for the period 1979–2014
- 2015**      **European Geosciences Union (EGU) General Assembly**  
12–17 abr 2015 · Vienna, Austria  
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings
- 2015**      **SPARC Regional Workshop — Role of the stratosphere in climate variability and prediction**  
12–13 ene 2015 · Granada, Spain  
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings

- 2010 Mexican Geophysical Union Annual Meeting**  
 7–12 nov 2010 · Puerto Vallarta, Mexico
- Seasonality and anomalies of oceanographic remote-sensing variables in the Gulf of Mexico
  - Seasonality and anomalies of sea surface height in the Caribbean Sea using altimetry data
  - An alternative for computationally intensive oceanographic data processing through parallel programming in MATLAB
- 2010 53rd Mexican National Physics Congress**  
 25–29 oct 2010 · Boca del Río, Veracruz, Mexico
- High-performance computing applied to parallel programming to optimise processes in Astrophysics and Physical Oceanography
- 2009 52nd Mexican National Physics Congress (SMF)**  
 26–30 oct 2009 · Acapulco, Guerrero, Mexico
- Frequency spectra of meteorological data from María Madre Island
  - Comparison of coastal and marine winds in the coastal zone of San Blas, Nayarit (1999–2002)

## ACTIVE RESEARCH PROJECTS

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- 2026 – present Flood early-warning system for the Guadalajara Metropolitan Area — Principal investigator**  
 Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding  
 Development of an urban flood early-warning system that combines remote sensing (IAM-UdeG weather radar), hydrological runoff modeling over high-resolution LiDAR terrain, and official flood inventories. The system classifies risk by sub-basin in real time and estimates water arrival time, aiming to calibrate thresholds through a retrospective analysis with historical Civil Protection/Fire Department records.
- 2026 – present Generative AI for lab practice reports (Google Gems) — Principal investigator**  
 Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding  
 Implementation of Google Gems as a writing and feedback assistant for students to draft and self-assess their lab practice reports in physics courses.
- 2025 – present AI-generated educational comics for science teaching — Principal investigator**  
 Universidad de Guadalajara · Funding: No external funding  
 Development and evaluation of educational comics created with generative AI tools as a didactic resource for teaching physics, mathematics and science at basic and upper-secondary levels.
- 2025 – present Pedagogical Calculus Assistant with Google Gems: development and student perception — Principal investigator**  
 Universidad Tecnológica de Jalisco · Funding: Universidad Tecnológica de Jalisco  
 Design and implementation of a Calculus learning assistant based on Google Gems, with quantitative and qualitative study of students' perception, satisfaction and academic performance.
- 2025 – 2028 Optical elements to enhance visual acuity via wavefront coding — Collaborator**  
 Universidad de Guadalajara · Funding: Universidad de Guadalajara  
 Design and validation of optical elements that improve visual acuity through wavefront coding.
- 2025 – present World Wide Lightning Location Network (WWLLN) — Collaborator · antenna lead**  
 University of Washington · Funding: University of Washington  
 Global lightning location network. In charge of the receiving antenna/node and real-time data delivery.
- 2024 – present Tlakakini Science & Technology Club — Coordinator**  
 CUCEI · Universidad de Guadalajara · Funding: University of Guadalajara · CUCEI  
 Science-outreach club for children, with monthly experiment and workshop sessions.

## TECH DEVELOPMENTS

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**Flood early-warning system · Guadalajara Metro Area**  
 Python · FastAPI · Leaflet · NumPy · pysheds · rasterio · GDAL · Docker  
 Real-time web platform that anticipates urban flooding in the Guadalajara Metropolitan Area. It fuses the IAM-UdeG weather radar (one scan per minute) with a hydrological runoff model built on 5 m LiDAR and Copernicus DEM terrain, plus the IMEPLAN official flood inventory (363 recurrent sites). It produces a per-zone risk light —calibrated against official water-depth records— and a continuous risk grid that routes rainfall downslope, with concentration time and a "water incoming" ETA. Includes live and historical-replay views. Deployed in production.

## COMMUNITY BUILDING & SERVICE

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Thesis supervision

**2011 Estacionalidad y anomalías del nivel del mar en el Caribe por medio de datos de altimetría**  
 Lilia Lizeth Sierra Romero · Bachelor's · Co-advisor · Universidad de Guadalajara

Journal peer review

- 2025** **IEEE Latin America Transactions**  
Reviewer — Q2 journal, double-blind
- 2024–2025** **Int. J. of Environment and Pollution (Inderscience)**  
Reviewer — Q4 journal, double-blind
- 2025** **Congreso de Ingeniería, Ciencia y Gestión Ambiental (AMICA)**  
Scientific committee reviewer
- Project evaluation**
- 2025–2026** **SECIHTI**  
External evaluator — 2026 National Call, Basic & Frontier Science (3 proposals)

## SCIENCE OUTREACH

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- 2024 –** **Tlakakini Science & Technology Club**  
Co-founder. Monthly outreach sessions for children on national non-school days.
- 2009 –** **Physics conferences (AGU, EGU, SMF)**  
Talks at national and international conferences on climate and the stratosphere.
- 2005 –** **Training science communicators**  
Optics and electromagnetism workshops; National Outreach Meeting.
- 2009 –** **Red Temática de Medio Ambiente y Sustentabilidad (Environment & Sustainability Thematic Network)**  
Member of the CONACYT environment and sustainability network.
- 2011 –** **Environmental Physics Laboratory (EPhysLab)**  
External collaborating researcher with the University of Vigo group.

## GRANTS & HONORS

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- 2025–2029** **National Researcher System · Candidate**  
SECIHTI · Mexico
- 2012–2016** **FPI Grant · Predoctoral Researcher**  
Ministerio de Educación · Spain
- 2011–2012** **Climate Science Master's Scholarship**  
Campus do Mar · Spain
- 2011–2012** **«Ourense Exterior» mobility grant**  
Diputación de Ourense · Spain
- 2009–2011** **Hydrometeorology Master's Scholarship**  
CONACYT · Mexico
- 2009** **Admission to the Sistema Estatal de Investigación (Jalisco State Research System)**  
COECYTJAL · Jalisco, Mexico
- 2009** **Bachelor's Thesis Scholarship**  
SEP · Mexico

## CONTINUING EDUCATION

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- 2025** **Diploma** — Techno-pedagogical strategies for developing 21st-century skills · Universidad de Guadalajara
- 2025** **Course** — Artificial Intelligence for teaching · Universidad de Guadalajara
- 2025** **Course** — Generative Artificial Intelligence for teachers · Universidad Anáhuac
- 2025** **Course** — Generative AI in the classroom · UNAM
- 2025** **Course** — Python for Data Science, AI & Development · IBM
- 2025** **Certification** — What is Data Science? · IBM
- 2025** **Course** — Differential Calculus for Data Science and AI · Platzi
- 2025** **Course** — Statistics and Probability · Platzi
- 2025** **Course** — Foundations of Software Engineering · Platzi
- 2025** **Course** — Introduction to the Terminal and Command Line · Platzi
- 2014** **Course** — OOP in Atmospheric and Oceanic Model Development · Universidade de Vigo
- 2011** **Diploma** — Teaching Competencies: Introduction to General High School Education · Universidad de Guadalajara

- 2005**      **Workshop** — Training for Science Communicators · Universidad de Guadalajara
- 2025**      **Course** — Mathematical Functions for Data Science and AI · Platzi
- 2005**      **Course** — Theoretical and Conceptual Development of Electromagnetism (Part I) · Universidad de Guadalajara
- 2009**      **Training** — DOCA Module I: The National System of Technological Education · Universidad de Guadalajara

## LANGUAGES

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Spanish — Native · English — Intermediate

## TOOLBOX

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Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter